

Nonlinear Control System-Assignment1-Xinyi Yang

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Problem 1

Consider the Lyapunov function

$$V(x) = \int_0^{x_1} h(\sigma) d\sigma + \frac{1}{2}(x_1 + x_2)^2, \quad h(x_1) = x_1 \left(2 + \frac{x_1^2}{1 + x_1^2} \right).$$

Positive definiteness and radial unboundedness. Since

$$x_1 h(x_1) = x_1^2 \left(2 + \frac{x_1^2}{1 + x_1^2} \right) \geq 2x_1^2, \quad \forall x_1,$$

it follows that

$$\int_0^{x_1} h(\sigma) d\sigma \geq \int_0^{x_1} 2\sigma d\sigma = x_1^2,$$

and hence $V(x) > 0$ for all $x \neq 0$. Moreover,

$$V(x) \geq x_1^2 + \frac{1}{2}(x_1 + x_2)^2 \rightarrow \infty \quad \text{as } \|x\| \rightarrow \infty,$$

so V is radially unbounded.

Time derivative. The time derivative of V along system trajectories is

$$\begin{aligned} \dot{V} &= h(x_1)\dot{x}_1 + (x_1 + x_2)(\dot{x}_1 + \dot{x}_2) \\ &= h(x_1)x_2 + (x_1 + x_2)(x_2 - h(x_1) - 2x_2) \\ &= -x_1 h(x_1) - (x_1 + x_2)^2. \end{aligned}$$

Using the inequality $x_1 h(x_1) \geq 2x_1^2$, we obtain

$$\dot{V} \leq -2x_1^2 - (x_1 + x_2)^2 \leq -c\|x\|^2$$

for some constant $c > 0$. Therefore, the origin is globally exponentially stable.

Problem 2

(a) Using the representation

$$f(x) = \int_0^1 \frac{\partial f}{\partial x}(\sigma x) x d\sigma,$$

we compute

$$x^T P f(x) + f^T(x) P x = \int_0^1 x^T (P J(\sigma x) + J^T(\sigma x) P) x d\sigma,$$

where $J(x) = \frac{\partial f}{\partial x}(x)$. By assumption,

$$P J(\sigma x) + J^T(\sigma x) P \leq -I,$$

for all $\sigma \in [0, 1]$, and hence

$$x^T P f(x) + f^T(x) P x \leq \int_0^1 (-x^T x) d\sigma = -x^T x.$$

(b) Let

$$V(x) = f^T(x) P f(x).$$

Then

$$\begin{aligned} \dot{V} &= \dot{f}^T P f + f^T P \dot{f} \\ &= f^T (P J + J^T P) f \\ &\leq -\|f(x)\|^2. \end{aligned}$$

Therefore, $\dot{V} \leq 0$ with equality only when $f(x) = 0$. Since $f(0) = 0$ and $f(x) \neq 0$ for $x \neq 0$, the largest invariant set contained in $\{\dot{V} = 0\}$ is $\{0\}$. By LaSalle's invariance principle, the origin is globally asymptotically stable. From part (a), we have $x^T P f(x) + f^T(x) P x \leq -x^T x$ for all $x \in \mathbb{R}^n$. This implies that $f(x)$ cannot be bounded away from zero for nonzero x , hence $V(x) = f^T(x) P f(x)$ is radially unbounded.

Problem 3

(a) Consider the Lyapunov function

$$V(x) = \int_0^{x_1} g_1(y) dy + \int_0^{x_2} g_2(y) dy + \frac{1}{2} x_3^2.$$

Since $y g_i(y) > 0$ for all $y \neq 0$, V is positive definite. The time derivative along trajectories is

$$\begin{aligned} \dot{V} &= g_1(x_1)(-x_1 + x_3) + g_2(x_2)x_3 + x_3(-g_1(x_1) - g_2(x_2) - x_3) \\ &= -x_1 g_1(x_1) - x_2 g_2(x_2) - x_3^2 < 0, \quad x \neq 0. \end{aligned}$$

Hence, the origin is asymptotically stable.

(b) If

$$\lim_{|y| \rightarrow \infty} \int_0^y g_i(s) ds = \infty, \quad i = 1, 2,$$

then V is radially unbounded and the origin is globally asymptotically stable.

(c) If there exists $c > 0$ such that

$$g_i(y)y \geq cy^2,$$

then

$$\dot{V} \leq -\alpha \|x\|^2$$

for some $\alpha > 0$, which implies exponential stability.

Problem 4

(a) At equilibrium, $x_2 = 0$ and

$$\sin x_1 + k_1 \text{sat}(x_1) = 0.$$

Since $k_1 > 1$, the only solution is $x_1 = 0$. Hence, the origin is the unique equilibrium.

(b) Consider

$$V(x) = \int_0^{x_1} (\sin \sigma + k_1 \text{sat}(\sigma)) d\sigma + \frac{1}{2}x_2^2.$$

Its derivative is

$$\dot{V} = -bx_2^2 - k_2x_2 \text{sat}(x_2) \leq 0.$$

By LaSalle's invariance principle, the origin is globally asymptotically stable.

Problem 5

Let

$$V(x) = \frac{1}{2}(x_1^2 + x_2^2).$$

Then

$$\dot{V} = x_1^4 - x_2^2.$$

The origin is asymptotically stable but not exponentially stable. Moreover, it is not globally asymptotically stable. An estimate of the region of attraction is

$$|x_1| < 1.$$

Problem 6

Consider

$$V(x) = \frac{1}{2}(x_1^2 + x_2^2 + x_3^2).$$

Along trajectories,

$$\dot{V} \leq -x_1^2 - x_2^2 - x_3^2 + 10|x_2||x_3|.$$

Hence, $\dot{V} < 0$ in a neighborhood of the origin, implying asymptotic stability. An estimate of the region of attraction can be obtained by bounding the cross term.

Problem 7

Consider the Lyapunov function

$$V(x) = \frac{x_1^2}{1+x_1^2} + x_2^2.$$

The gradient of V is

$$\nabla V(x) = \begin{bmatrix} \frac{2x_1}{(1+x_1^2)^2} \\ 2x_2 \end{bmatrix}.$$

The time derivative of V along system trajectories is

$$\begin{aligned} \dot{V} &= \langle \nabla V(x), f(x) \rangle \\ &= \frac{2x_1}{(1+x_1^2)^2} \frac{2x_2-6x_1}{(1+x_1^2)^2} + 2x_2 \frac{-2(x_1+x_2)}{(1+x_1^2)^2}. \end{aligned}$$

After simplification,

$$\dot{V} = -\frac{4x_1^2}{(1+x_1^2)^2} - \frac{4x_2^2}{(1+x_1^2)^2} < 0, \quad \forall x \neq 0.$$

Therefore, the origin is asymptotically stable. However, since the state space is partitioned by invariant curves, the origin is not globally asymptotically stable.

Problem 8

Let (x^*, ν^*) satisfy

$$\nabla f(x^*) + A^T \nu^* = 0, \quad Ax^* - b = 0.$$

Consider the Lyapunov function

$$V(x, \nu) = \frac{1}{2} \|x - x^*\|^2 + \frac{1}{2} \|\nu - \nu^*\|^2.$$

The derivative of V along trajectories is

$$\begin{aligned} \dot{V} &= (x - x^*)^T (-\nabla f(x) - A^T \nu) + (\nu - \nu^*)^T (Ax - b) \\ &= -(x - x^*)^T (\nabla f(x) - \nabla f(x^*)). \end{aligned}$$

Since f is strictly convex, we have

$$(x - x^*)^T (\nabla f(x) - \nabla f(x^*)) > 0, \quad x \neq x^*,$$

which implies

$$\dot{V} \leq 0.$$

Moreover, $\dot{V} = 0$ implies $x = x^*$, and from the system equations it follows that $\nu = \nu^*$. Thus, the largest invariant set contained in $\{\dot{V} = 0\}$ is (x^*, ν^*) .

By LaSalle's invariance principle, the equilibrium (x^*, ν^*) is globally asymptotically stable.